



## ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

### Section A: Submission Details

**Programme** : Bachelor in Information Technology (Honours) (Internet of Things)  
**Course Code & Name** : IIB43203 - CLOUD COMPUTING  
**Course Lecturer(s)** : Mohd Asyraf Abdul Rahman  
**Submission Title** : GROUP PROJECT  
**Deadline** : Day 7 Month 6 Year 2026 Time 11.59PM  
**Penalties** :

- 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted.
- Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

### Section B: Academic Integrity

Tick (✓) each box below if you agree:

- |   |                                                                                                |
|---|------------------------------------------------------------------------------------------------|
| / | I have read and understood the UniKL's policy on Plagiarism in University Rules & Regulations. |
| / | This submission is my own, unless indicated with proper referencing.                           |
| / | This submission has not been previously submitted or published.                                |
| / | This submission follows the requirements stated in the course.                                 |

### Section C: Submission Receipt

(must be filled in manually)

#### Office Receipt of Submission

| Date & Time of Submission (stamp) | Student Name(s)                                                    | Student ID(s)              |
|-----------------------------------|--------------------------------------------------------------------|----------------------------|
|                                   | MUHAMMAD ASYRAF WAJDI BIN MUSTAFFA<br>AHMAD SYAZWAN BIN SHAHRUDDIN | 52227124873<br>52227124582 |

---

#### Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

| Date & Time of Submission (stamp) | Course Code | Submission Title | Student ID(s)              |
|-----------------------------------|-------------|------------------|----------------------------|
|                                   |             |                  | 52227124873<br>52227124582 |



**MALAYSIAN INSTITUTE OF INFORMATION TECHNOLOGY**

**MARCH 2026 SEMESTER**

**CLOUD COMPUTING (IIB43203)**

**GROUP PROJECT**

**PREPARED BY:**

| <b>NO.</b> | <b>NAME</b>                       | <b>STUDENT ID</b> |
|------------|-----------------------------------|-------------------|
| 1.         | MUHAMMAD ASYRAF WAJD BIN MUSTAFFA | 52227124873       |
| 2.         | AHMAD SYAZWAN BIN SHAHRUDDIN      | 52227124582       |

**SUBMISSION ON DATE:**  
**7<sup>TH</sup> JUNE 2026**

## A. Amazon S3 Storage Setup

### 1. Bucket setup

#### 1.1 Open AWS console and choose S3, then click create bucket

**Default encryption** [info](#)  
Server-side encryption is automatically applied to new objects stored in this bucket.

**Encryption type** [info](#)  
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the Storage tab of the [Amazon S3 pricing page](#).

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

**Bucket Key**  
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

☒ Enable

► **Advanced settings**

1 After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

[Cancel](#) [Create bucket](#)

#### 1.2 Choose General purpose for General configuration, and set the bucket name to student-registration-system. Click Create Bucket.

**Create bucket** [info](#)  
Buckets are containers for data stored in S3.

**General configuration**

**AWS Region**  
US East (N. Virginia) us-east-1

**Bucket type** [info](#)

- ☒ **General purpose**  
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
- ☐ **Directory**  
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

**Bucket namespace**  
Choose the namespace where you want to create your bucket. [Learn more](#)

- ☒ **Global namespace**  
By default, S3 creates general purpose buckets in the global namespace.
- ☐ **Account Regional namespace (recommended)**  
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

**Bucket name** [info](#)  
student-registration-asywan

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

### 2. Uploading Media

#### 2.1 After bucket successfully created, select the new bucket and click upload.

Successfully created bucket "student-registration-asywan"

**student-registration-asywan** [info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

**Objects (0)** [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

| Name                                       | Type | Last modified | Size | Storage class |
|--------------------------------------------|------|---------------|------|---------------|
| No objects                                 |      |               |      |               |
| You don't have any objects in this bucket. |      |               |      |               |

[Upload](#)

## 2.2 Add file of the image and click Upload.

The screenshot shows the AWS S3 Upload interface. At the top, there's a section titled "Upload" with an "Info" link. Below it, a message says: "Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)". A dashed box indicates where to drag and drop files. Below this, a section titled "Files and folders (1 total, 12.7 KB)" shows a table with one file: "Student Thumb up.jpeg" of type "image/jpeg" and size "12.7 KB". There are "Remove", "Add files", and "Add folder" buttons. Below the table, there's a "Destination" section with an "Info" link. It shows the destination as "s3://student-registration-asywan" and includes expandable sections for "Destination details", "Permissions", and "Properties". At the bottom right, there are "Cancel" and "Upload" buttons.

**Upload** Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

**Files and folders** (1 total, 12.7 KB)

All files and folders in this table will be uploaded.

| <input type="checkbox"/> | Name                  | Folder | Type       | Size    |
|--------------------------|-----------------------|--------|------------|---------|
| <input type="checkbox"/> | Student Thumb up.jpeg | -      | image/jpeg | 12.7 KB |

[Remove](#) [Add files](#) [Add folder](#)

**Destination** Info

**Destination**

[s3://student-registration-asywan](#)

► **Destination details**

Bucket settings that impact new objects stored in the specified destination.

► **Permissions**

Grant public access and access to other AWS accounts.

► **Properties**

Specify storage class, encryption settings, tags, and more.

[Cancel](#) [Upload](#)

🟢 **Upload succeeded**

For more information, see the [Files and folders](#) table.

**Upload: status** [Close](#)

ⓘ After you navigate away from this page, the following information is no longer available.

**Summary**

|                                  |                             |                     |
|----------------------------------|-----------------------------|---------------------|
| <b>Destination</b>               | <b>Succeeded</b>            | <b>Failed</b>       |
| s3://student-registration-asywan | 🟢 1 file, 12.7 KB (100.00%) | 🔴 0 files, 0 B (0%) |

[Files and folders](#) [Configuration](#)

**Files and folders** (1 total, 12.7 KB)

| Name                                  | Folder | Type       | Size    | Status      | Error |
|---------------------------------------|--------|------------|---------|-------------|-------|
| <a href="#">Student Thumb up.jpeg</a> | -      | image/jpeg | 12.7 KB | 🟢 Succeeded | -     |

## B. AWS EC2 setup

### 1. Launch instance

1.1 in AWS console select EC2, then click Launch instance.

The screenshot shows the AWS Management Console for the EC2 service. On the left, there's a navigation menu with "EC2" selected, and sub-items like "Dashboard", "AWS Global View", "Events", "Instances", "Instance Types", "Launch Templates", "Spot Requests", "Savings Plans", "Reserved Instances", "Dedicated Hosts", "Capacity Reservations", and "Capacity Manager". The main content area is titled "Resources" and says: "You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:". Below this, there's a table showing resource counts: "Instances (running)" (1), "Dedicated Hosts" (0), "Key pairs" (2), "Security groups" (10), "Auto Scaling Groups", "Elastic IPs", "Load balancers", and "Snapshots". At the bottom, there's a "Launch instance" section with the text: "To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud." and two buttons: "Launch instance" and "Migrate a server".

**EC2**

[Dashboard](#)

[AWS Global View](#)

[Events](#)

▼ **Instances**

[Instances](#)

[Instance Types](#)

[Launch Templates](#)

[Spot Requests](#)

[Savings Plans](#)

[Reserved Instances](#)

[Dedicated Hosts](#)

[Capacity Reservations](#)

[Capacity Manager](#) **New**

**Resources**

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

|                                     |    |                                     |
|-------------------------------------|----|-------------------------------------|
| <a href="#">Instances (running)</a> | 1  | <a href="#">Auto Scaling Groups</a> |
| <a href="#">Dedicated Hosts</a>     | 0  | <a href="#">Elastic IPs</a>         |
| <a href="#">Key pairs</a>           | 2  | <a href="#">Load balancers</a>      |
| <a href="#">Security groups</a>     | 10 | <a href="#">Snapshots</a>           |

**Launch instance**

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

[Launch instance](#) [Migrate a server](#)

## 1.2 Write the name and tags

**Launch an instance** [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

**Name and tags** [Info](#)

Name

student-registration-server-asywan

[Add additional tags](#)

## 1.3 For the OS, choose Ubuntu.

**▼ Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

**Recents** **Quick Start**

Amazon Linux macOS **Ubuntu** Windows Red Hat SUSE Linux Debian

**Amazon Machine Image (AMI)**

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type Free tier eligible

ami-05cf1e9f73fbad2e2 (64-bit (x86)) / ami-0953e222326856ce (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

**Description**

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

| Architecture | AMI ID                | Publish Date | Username |                                   |
|--------------|-----------------------|--------------|----------|-----------------------------------|
| 64-bit (x86) | ami-05cf1e9f73fbad2e2 | 2026-04-24   | ubuntu   | <a href="#">Verified provider</a> |

## 1.4 On Create key pair, write the key pair name and choose RSA key pair type. Keep the private key file format in .pem. Click Create key pair.

**Create key pair**

**Key pair name**

Key pairs allow you to connect to your instance securely.

studentkey

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

**Key pair type**

☒ **RSA**  
RSA encrypted private and public key pair

☐ **ED25519**  
ED25519 encrypted private and public key pair

**Private key file format**

☒ **.pem**  
For use with OpenSSH

☐ **.ppk**  
For use with PuTTY

**When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)**

[Cancel](#) [Create key pair](#)

**1.5** For network setting, choose Create security group on Firewall (security groups). Tick Allow SSH and Allow HTTP. Make sure to Click Launch instance.

The screenshot shows the AWS Management Console 'Network settings' page. Under 'Firewall (security groups)', the 'Create security group' option is selected. Below, it states: 'We'll create a new security group called 'launch-wizard-2' with the following rules:'. Two rules are checked: 'Allow SSH traffic from' (source: Anywhere, 0.0.0.0/0) and 'Allow HTTP traffic from the Internet'. The 'Summary' panel on the right shows 1 instance, Canonical Ubuntu 24.04 AMI, t3.micro instance type, and 1 volume (8 GiB). The 'Launch instance' button is visible.

## 2. Setup Apache and default page

**2.1** open terminal by select our EC2 instance and click connect

| Name              | Instance ID         | Instance state | Instance type | Status check      | Availability Zone | Public IPv4 DNS          | Public IPv4 ... | Elastic IP | IPv6 IPs | Monitoring | Security |
|-------------------|---------------------|----------------|---------------|-------------------|-------------------|--------------------------|-----------------|------------|----------|------------|----------|
| Asyraf-ec2        | i-069c361b4e1aae2f4 | Running        | t3.micro      | 3/3 checks passed | us-east-1a        | ec2-100-52-193-247.co... | 100.52.193.247  | -          | -        | disabled   | ec2-rds  |
| student-regist... | i-07a6e02a7246facaf | Running        | t3.micro      | 3/3 checks passed | us-east-1d        | ec2-3-81-151-210.com...  | 3.81.151.210    | -          | -        | disabled   | ec2-rds  |

**2.2** install Apache

```
ubuntu@ip-172-31-32-124:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
```

**2.3** start and enable Apache

```
ubuntu@ip-172-31-32-124:~$ sudo systemctl start apache2
ubuntu@ip-172-31-32-124:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
```

**2.4** Edit the web page html

```
ubuntu@ip-172-31-32-124:~$ cd /var/www/html
ubuntu@ip-172-31-32-124:/var/www/html$ ls
index.html
ubuntu@ip-172-31-32-124:/var/www/html$ sudo rm index.html
ubuntu@ip-172-31-32-124:/var/www/html$ sudo nano index.html
```

## 2.5 Put the html code for the web page.

```
GNU nano 7.2
<!DOCTYPE html>
<html>
<head>
<title>Student Registration System</title>
</head>
<body>
<h1>Student Registration System</h1>
<p>Welcome to my AWS Project.</p>
</body>
</html>
```

## 3. Install PHP and setup registration form

### 3.1 install the PHP in the terminal and edit the register.php

```
ubuntu@ip-172-31-32-124:/var/www/html$ cd /var/www/html
ubuntu@ip-172-31-32-124:/var/www/html$ sudo nano register.php
```

### 3.2 Put the code of the registration form

```
GNU nano 7.2
<!DOCTYPE html>

<html>
<head>
    <title>Student Registration System</title>
</head>
<body>

<h1>Student Registration System</h1>

<form method="post">
...
<label>Full Name:</label><br>
<input type="text" name="fullname" required><br><br>

<label>Email Address:</label><br>
<input type="email" name="email" required><br><br>

<label>Phone Number:</label><br>
<input type="text" name="phone" required><br><br>

<input type="submit" value="Register">
...

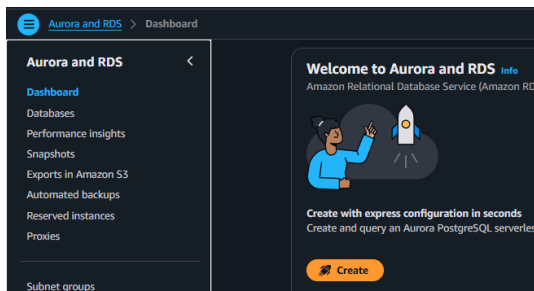
</form>

</body>
</html>
```

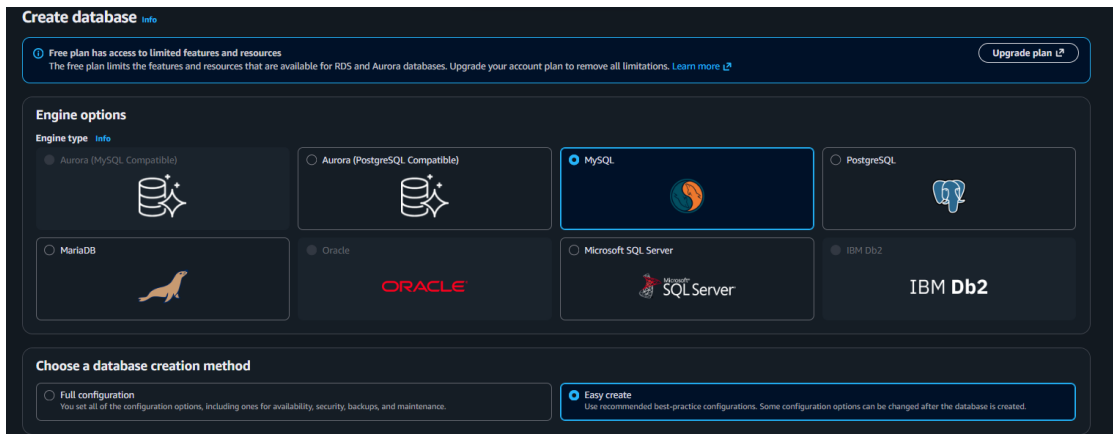
## C. Amazon RDS Database Setup

### 1. Manage the database

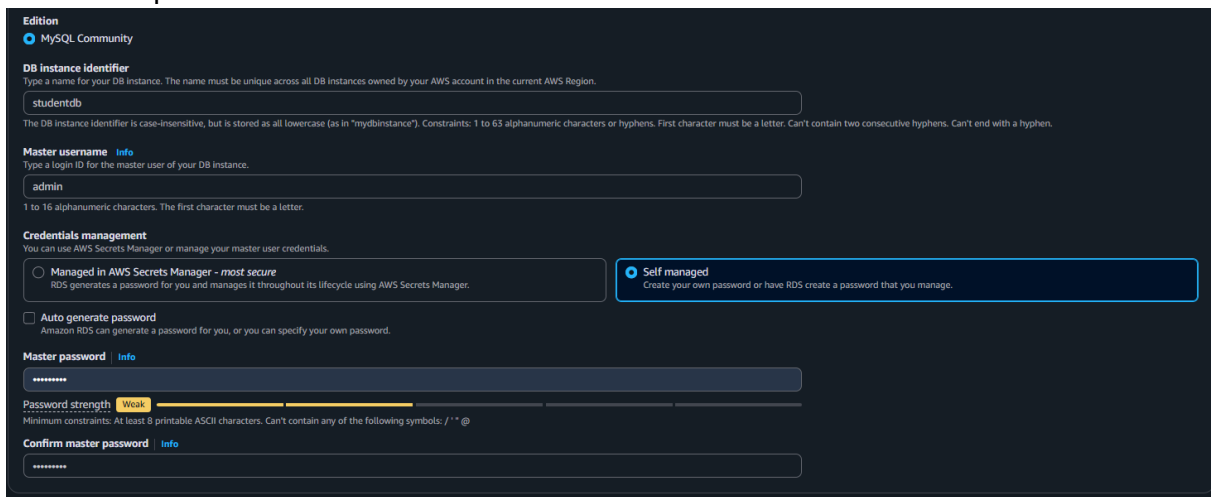
#### 1.1 Click Create



#### 1.2 Choose MySQL for Engine options and choose Easy create for database creation method.



#### 1.3 Write the studentdb in DB instance identifier. Choose Self managed and create the Master password.





## 1.4 Select Connect to an EC2 compute resource to connect to our EC2 instance earlier. Click Create database.

**▼ Set up EC2 connection - optional**  
You can also set up a connection to an EC2 instance after creating the database. Go to the database list page or the database details page, choose Actions, and then choose Set up to EC2 connection.

**Compute resource**  
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource  
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ **Connect to an EC2 compute resource**  
Set up a connection to an EC2 compute resource for this database.

**EC2 instance** Info  
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-07a6e02a7246facaf  
student-registration-server-asywan

ⓘ You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

Cancel Create database

## 2. Connect EC2 to MySQL

### 2.1 Open the terminal of EC2 and install MySQL

```
ubuntu@ip-172-31-32-124:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
40 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-32-124:~$ sudo apt install mysql-client -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mysql-client is already the newest version (8.0.46-0ubuntu0.24.04.2).
0 upgraded, 0 newly installed, 0 to remove and 40 not upgraded.
```

### 2.2 Open MySQL prompt and create database table.

```
ubuntu@ip-172-31-32-124:~$ mysql -h studentdb.ce3kimoi0r74.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 44
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE student_registration;
Query OK, 1 row affected (0.03 sec)

mysql> USE student_registration;
Database changed
mysql> CREATE TABLE students (
->   id INT AUTO_INCREMENT PRIMARY KEY,
->   fullname VARCHAR(100),
->   email VARCHAR(100),
->   phone VARCHAR(20)
-> );
Query OK, 0 rows affected (0.12 sec)
```

### 2.3 Edit register.php again and put the code to connect the website with database.

```
GNU nano 7.2
<?php
$host = "studentdb.ce3kimoi0r74.us-east-1.rds.amazonaws.com";
$user = "admin";
$password = "Admin123!";
$database = "student_registration";

$conn = new mysqli($host, $user, $password, $database);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$message = "";

if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $fullname = $_POST['fullname'];
    $email = $_POST['email'];
    $phone = $_POST['phone'];

    $sql = "INSERT INTO students (fullname, email, phone)
            VALUES ('$fullname', '$email', '$phone')";

    if ($conn->query($sql) === TRUE) {
        $message = "Student registered successfully!";
    } else {
        $message = "Error: " . $conn->error;
    }
}

?>
```

```

<!DOCTYPE html>

<html>
<head>
    <title>Student Registration System</title>
</head>
<body>

<h1>Student Registration System</h1>

<p><?php echo $message; ?></p>

<form method="post">
    ...
    <label>Full Name:</label><br>
    <input type="text" name="fullname" required><br><br>

    <label>Email Address:</label><br>
    <input type="email" name="email" required><br><br>

    <label>Phone Number:</label><br>
    <input type="text" name="phone" required><br><br>

    <input type="submit" value="Register">
    ...

</form>

</body>
</html>

```

#### D. Web test

##### 1. Web registration

###### 1.1 fill the form

## Student Registration System

Full Name:

Email Address:

Phone Number:

1.2 data registered

# Student Registration System

Student registered successfully!

Full Name:

Email Address:

Phone Number:

Register

## 2. Check the data in terminal

```
mysql> SELECT * FROM students;
+----+-----+-----+-----+
| id | fullname | email | phone |
+----+-----+-----+-----+
| 1 | syazwan | student@unikl | 0123456789 |
+----+-----+-----+-----+
1 row in set (0.00 sec)
```

Youtube Presentation Link

[https://youtu.be/\\_7cLHPSR8RI?si=Adrd5OaQL74Kzwxx](https://youtu.be/_7cLHPSR8RI?si=Adrd5OaQL74Kzwxx)